Understanding, Anthropocentrically Defined:

What the Competence-Performance Gap in Machine Reasoning Reveals about a Contested Concept

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1. Motivation: knowledge versus understanding

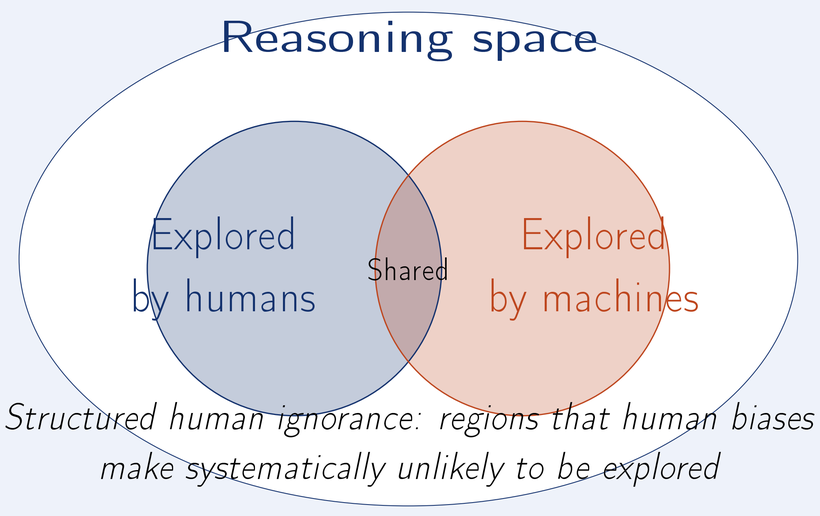
Contemporary epistemology increasingly argues that **understanding**, not knowledge, is the more demanding and more central epistemic achievement (Grimm, 2012; Elgin, 2017; Zagzebski, 2001).

- **Knowledge** concerns individual true beliefs.
- **Understanding** requires grasping how the elements of a domain depend on and relate to one another.

Question. What does this distinction mean once the class of possible "graspers" is no longer confined to human minds?
We use the discovery of unexplored reasoning routes in artificial systems as a case study.

2. The reasoning space

A **reasoning space** is the set of possible trajectories of hypotheses, abstractions, and actions available when addressing a problem.



Structured human ignorance: regions that human biases make systematically unlikely to be explored

A human solver explores only a limited portion of this space, shaped by inductive biases, perceptual habits, and recurring patterns of premature commitment to a hypothesis. A machine reasoning system explores its own, generally different, portion.

3. Human ignorance as structure

We characterise human ignorance **not as a lack of information** but as a structured region of the reasoning space that human biases make systematically unlikely to be explored.

This holds **even when all the facts needed to reach a solution are already available**.

4. Testbed: ARC

We use the **Abstraction and Reasoning Corpus (ARC)** as a controlled testbed for comparing human and machine reasoning.

We ask what it would take for a system that reaches into the unexplored region to be said to *understand* something, rather than merely to *perform well*.

5. Central argument

Reaching a single solution is structurally analogous to acquiring a piece of knowledge. Grasping the overall shape of the reasoning space is analogous to understanding in the fuller, philosophically demanding sense.

Reasoning space	Epistemic analogue
Reaching a single solution	Knowledge
Grasping the shape of the space, including where and why human exploration stops short	Understanding

6. Superior performance is not evidence of understanding

A system can be superhuman on a shared task through brute-force search or specialised heuristics while grasping nothing about *why* humans fail where they fail.

This is a structural cousin of Searle's (1980) Chinese Room worry, applied not to symbol manipulation but to the **inference from successful exploration of a reasoning space to genuine understanding of that space**.

7. A second-order criterion

A stronger candidate for understanding is a **second-order capacity**: not merely reaching a solution humans miss, but **predicting, explaining, and reasoning about why humans miss it**.

This mirrors what psychologists call **metacognition** (Flavell, 1979).

8. A residual problem for human evaluators

Drawing on Nagel's (1974) discussion of the limits of understanding a perspective from outside, even a system satisfying the second-order criterion poses a residual problem.

Human evaluators may be unable to verify, from their own cognitive vantage point, whether a machine's success reflects **grasped structure** or merely **opaque, successful computation**.

Conclusion

The philosophically interesting question raised by increasingly capable AI systems is **not whether they can solve problems humans cannot**, but whether the concept of understanding itself, developed with only human and biological cognisers in view, **needs to be reconceived** once it is asked to do work outside that domain.